

Tyler Nielsen

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SUMMARY

Computer Systems Engineering graduate with hands-on experience in embedded systems, hardware-based projects, and digital circuit design. Skilled in firmware development, communication protocols and real-time operating systems, with project work integrating and testing embedded software on, and communicating with, electronics systems. Eager to apply engineering ability and problem-solving skills to building and validating embedded technologies.

EDUCATION

Bachelor of Science in Engineering, Computer Systems Engineering

May 2026

Arizona State University, Tempe, AZ

4.0 GPA

Relevant Coursework: Embedded Microprocessor Systems, Distributed Software Development, Operating Systems, Reconfigurable Computing, Computer Architecture, Foundations of Machine Learning

SKILLS

Embedded Systems, Linux, Git/GitHub, Jira, pytest, FreeRTOS, Pico C/C++ SDK, Project Management, Peer Tutoring, Technical Writing, Vivado, ARM Cortex, KiCad, PCB Integration, ASP.NET

Languages: C, C++, Python, Java, C#, Verilog

PROFESSIONAL EXPERIENCE

Tutor.com: Intermediate Tutor

February 2024 – Present

- Tutor undergraduates in math and computer science, including C, C++, Java, Python, Calculus, and Linear Algebra.
- Maintain a 4.7/5 student rating for over 1000 sessions.

Arizona Space Grant Consortium: NASA Space Grant Intern

August 2025 – May 2026

- Designed modular space software and hardware stack to streamline development of space and high-altitude oriented projects for the Interplanetary Laboratory at Arizona State University.
- Outlined and performed High-Dose Gamma Radiation stress testing of the Raspberry PI RP-Series microcontrollers, publishing findings to the USU Small Satellite Conference.
- Repaired the Lab's existing UHF satellite ground stations and built a new backup to the primary TX/RX station, with both TX setups capable of delivering a clean 50W signal with LoRa modulation support.

Arizona State University: Undergraduate Teaching Assistant

August 2025 – December 2025

- Worked as part of the staff team for CSE 320 Fundamentals of Digital Design. Held office hours and aided students in FPGA development with Verilog using Vivado Design Suite on Artix 7 development boards.

Brigham Young University: Undergraduate Research Assistant

April 2023 – July 2023

- Implemented Python scripts to run satellite footprint simulations for the GLOWS satellite from BYU's Microwave Earth Remote Sensing Lab using custom Python libraries.

PROJECT EXPERIENCE

General Dynamics Missions Systems Smart Inventory Tracker

Fall 2025 – Spring 2026

Backend Developer - Worked on a team of 5 following Agile methodologies using Jira to build a hardware-integrated, locally-hosted ASP.NET web application using .NET 10 to enable asset tracking via QR codes and/or RFID:

- Completed a trade study and designed the system to enable both cost effective QR and more efficient RFID technologies to be used interchangeably while integrating legacy inventory records.
- Used Aspire AppHost to orchestrate backend resources and PostgreSQL database integration.
- Designed secure RESTful API endpoints to enable front-end functionality with JWT authentication, EF Core database integration, and user management systems.

Sun Devil Satellite Laboratory Coconut CubeSat

Spring 2024 – Summer 2026

Software Team Lead - Led a team of 7 to create and test embedded flight software on an ARM Cortex-based microcontroller for a store-and-forward satellite using FreeRTOS and the Pico C/C++ SDK, deployed to orbit in July 2026:

- Integrated flight software with the electronic system using multicore SWD debugging and electronics analysis tools.
- Developed a LoRa radio task, packet definitions, and physical interfaces including I2C drivers and USB management.
- Created a Python test suite utilizing OpenC3 COSMOS for automation and configuration for Day-In-The-Life testing.

Arizona State University ASCEND High-Altitude Balloon

Spring 2024 – Spring 2026

Software Team Lead - Led a team of 10 to develop embedded, dual-core, modular flight software in C++ alongside multithreaded ground software in Python to provide access to real-time telemetry data for a high-altitude balloon payload:

- Designed a custom packet system, reducing needed onboard storage and telemetry data rate by 67%.
- Utilized Taiga and GitHub for collaboration, progress tracking, and maintenance automation with GitHub Actions.
- Developed a multithreaded ground station GUI with Tkinter with a pipelined packet decoding and storage system.